

Meeting Summaries of March 19

Date: March 19

Time: 18:05 - 19:10 PM(PDT)

Attendees: Prof. Jan, Dr. Javed, Dr. Yue and Feng

Topic: Discussion of the work of **Chapter 2:** Stressors, Contamination Scores and amount of least polluted sites; Advices on the **RDA analysis** in picking the cut-off value for the least polluted sites; The confirmation of continue work on **Chapter 3**

Feedbacks, Questions and Suggestions

- Dr. Javed: Use **simple language**, not in math, to explain what you did and obtain in each step. About the **cut-off value** selection of RDA, is it only using the $\bar{R}^2$ value to select the optimal cut-off value? Or did Feng use more criteria beyond the $\bar{R}^2$ value to pick out the optimal cut-off value?
- Prof. Jan: Is there a minimum sample sizes you should need for any of these statistical procedures in the RDA supported process of finding optimal cut-off point? For example, would this 25% proportion be valid for a total 4 or 10 samples? Is there a **rule of thumb** of this multivariate analysis? Is it possible to clarify which amount is minimum need and safe to use among the $n = 10, 20$ or 30 scenarios?
- Dr. Javed and Prof. Jan: This RDA analysis is good in comparing ecological views and selecting the cut-off point for least polluted sites from all contaminantly assessed sites. **But** it is worth more plain and precise words to explain the ideas, steps and results. How the results then validate the ideas that you presented, is it good and how confidently would you use it and move forwards?
- Dr. Javed and Prof. Jan: Write couple of paragraphs with plain words to interpret the RDA validation figure, the unusual, surprising and confusing parts in the figure need to be focus on. For example, why the two views gave least polluted sites with very different RDA performance at the beginning, does it mean the **MaxRel** view almost failed to capture the environment-community structure with the least polluted sites? Then, how do the RDA performance of the two views change across the candidate cut-off values and why they finally converge at the end?
- Prof. Jan: When it is using environment features to predict the taxa features without any contamination stressors at all, what do the RDA actually work for and how latently do the contamination stressors affect that relationship?

Not assuming contamination gone but quantifying it

Think of the contamination stressors with some ideas that could quantify their latent effects could be good, instead of only assuming their effects are negligible in the confined least polluted sites.

- Prof. Jan: Why was the $\bar{R}^2$ **negative** of the MaxRel view at the beginning? How negative could a regression's performance be? What is the **rule of thumb** of stating a good prediction model? How much variation you expect to see in a typical regression model in a similar context as you did? Can you make therefore a comparison of the model performance of your's?
- Prof. Jan: How the RDA's performance on **a randomization scenario**? For example, if you randomly select the least polluted sites, how would the RDA perform? This could be **a benchmark** to compare your RDA performance with the selected least polluted sites than doing no comparison at all.
- Prof. Jan: Another view or question Feng can think of is that, 'How representative are the contamination stressors of the samples for the whole corridor?' A context of this question is that, the whole corridor can hold numerous sites, but limited sources support only limited samples being collected and analyzed. Regarding the **estimates** from the samples, how confident or how similar are the estimates from these specific samples to represent the **true parameters** of the whole corridor that could generate much more unknown samples?

Min-max scaling vulnerability to unknown but existing extremes

Specifically, regarding the **min-max** scaling of the stressor scores, although the minimum (0) and maximum (1) scales are useful to sort out a comparably equal range of the stressor scores. But, this comparability is more rooted in the **relative ranking of the collected sites** and could be distorted by newly collected sites with more extreme values than the current minimum and maximum. Therefore be cautioned about the **strong local standardization** effect of the min-max scaling over the samples, which may lead its downstream analysis less representative and robust when more extreme sites indeed exist in the real world.

- Prof. Jan: The continuity of finishing the thesis is still primary. Feng can carefully pick justified results and move on to chapter 3. Regarding the new findings and confusions in this chapter 2, the **attached articles** in the email gave some information about how the contamination levels kill the animals and can be supportive to argue where the limit of contamination level should stop for preserving the least polluted sites.

Min-max scaling vulnerability to unknown but existing extremes

If the attached article discussed the correlation between the contamination levels and the animal mortality in a similar context, then it is the part that my RDA is to include. This part would elucidate the role of contamination stressors in the RDA process to help determine the optimal cut-off value, corresponding to the reflection - **Not assuming contamination gone but quantifying it**.

Summary of the Work in Chapter 2

- Feng gave a report on the work of **Chapter 2**, it consists of **three objectives** and the major steps and results were presented. They were presented in the following order and ways:
 - Objective 1: Extract **latent contamination stressors** from the original multivariate chemical data. The major steps are as follows:
 1. Log-transform and standardize the original chemical data to eliminate extreme values and remove unintended variation weights in PC loadings, it produced the Z matrix as the processed chemical data.
 2. Perform **PCA** on the Z matrix to extract and retain PCs that correspond to the latent contamination **stressors**, **sorting and analyzing** the variable loadings of the stressors to discuss the structure within each stressor and the relationships between the stressors.
 3. Implement sign orientation if needed to make sure *'higher stressor values indicate higher contamination levels'*. Then, apply **min-max scaling** to the stressor scores by holding the assumption that all stressors are equal in toxicity and distorting the ecological community.
 4. Finalize the **stressor matrix** S , each column represents a stressor with its values fall in the range of $[0, 1]$, the higher the value, the more contaminated on this stressor direction the site is.
 - Objective 2: Compare and pick out an **ecological view** to combine the stressors into one site-level contamination score. The major steps are as follows:
 1. On top of the stressor matrix S , calculate the candidate site-level contamination scores by applying different **ecological views**.
 - (a) **SumRel - cumulative-stress view**: sum up the stressor scores for each site, an ecological community should mainly be shaped by all the stressors but not a dominant one.
 - (b) **MaxRel - dominant-stress view**: take the maximum stressor score for each site, an ecological community should mainly be shaped by the dominant stressor.
 2. Discuss the reasons for why both views could be ecologically meaningful.
 3. Pick out the same number of least polluted sites based on the two views, then introduce the **RDA** to test which view preserves more constrained variation in the taxa data. The **constrained variation** is the total variation in the predicted taxa matrix Y' by the RDA model with environmental predictors, and the Y' is determined by not only the data set but also the underlying model and algorithm in fitting this **environment-community structure**.
 4. Acrossing several candidate amount of least polluted sites of the two views, the view always with better RDA performance indicates a **better preservation** of the natural **environment-community structure** and thus is more suitable for constructing the site-level contamination score for this study.
 - Objective 3: Determine the **cut-off** value (a percentile) for the least polluted sites.
 1. Start from the selected ecological view in Objective 2, prepare **candidate cut-off values** (percentiles) to split the sites into candidate least polluted set and the rest.

2. For each candidate least polluted site set, perform the RDA with its environment and taxa data, record the R^2 **value of the RDA**, the higher the R^2 value, the stronger the environment-community relationship being fitted by the RDA model.
3. Line plot the R^2 values across the candidate cut-off values, and pick out the cut-off value with a good R^2 value and stable neighbouring R^2 values as the **optimal cut-off value**. Then, the least polluted sites are determined by splitting the sites with this **optimal cut-off value**.

At the reaching of here, the **three objectives of Chapter 2** have been discussed. They are summarized as: a) identify latent contamination stressors; b) define the site-level contamination score; c) select the least polluted reference sites.

Action Items

- Feng will write a couple of paragraphs to **explain the RDA validation figure** with plain words, and focus on the unusual and confusing parts.
- Feng will flag and **complement the RDA work** based on the feedbacks and suggestions, for example, the randomization test of RDA performance and the quantifying contamination stressors' role in the RDA process.
- Feng will continue to **move on to Chapter 3** with a temporary results of Chapter 2, picking a conventional values for the cut-off value and implement the invertebrate assembly analysis.

Follow-Up

Temporary meeting time: March 26, 2025, 18:00 - 19:00 PM(PDT)

Supportive materials

- A meeting recording has been restored on Feng's local drive.